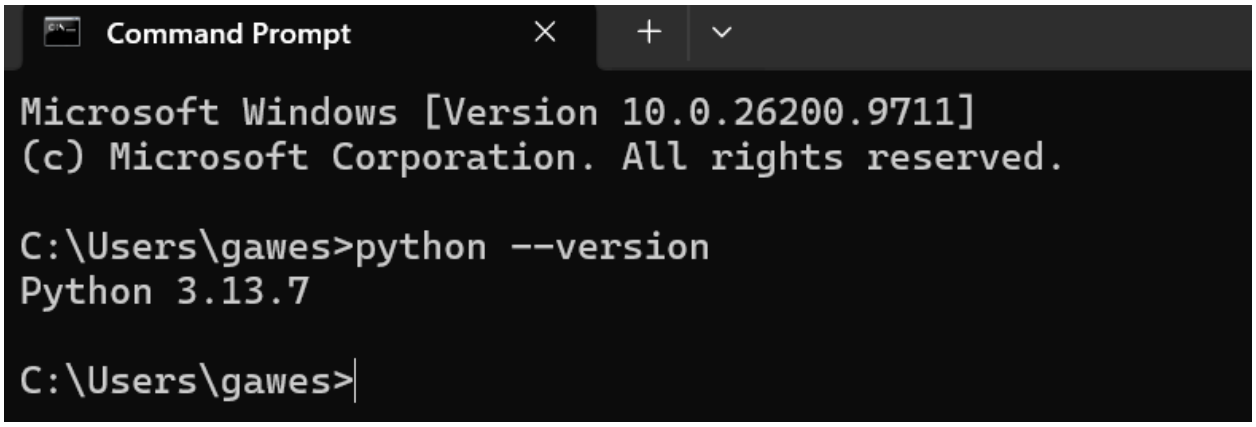
Lab 4 AI

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Step 1

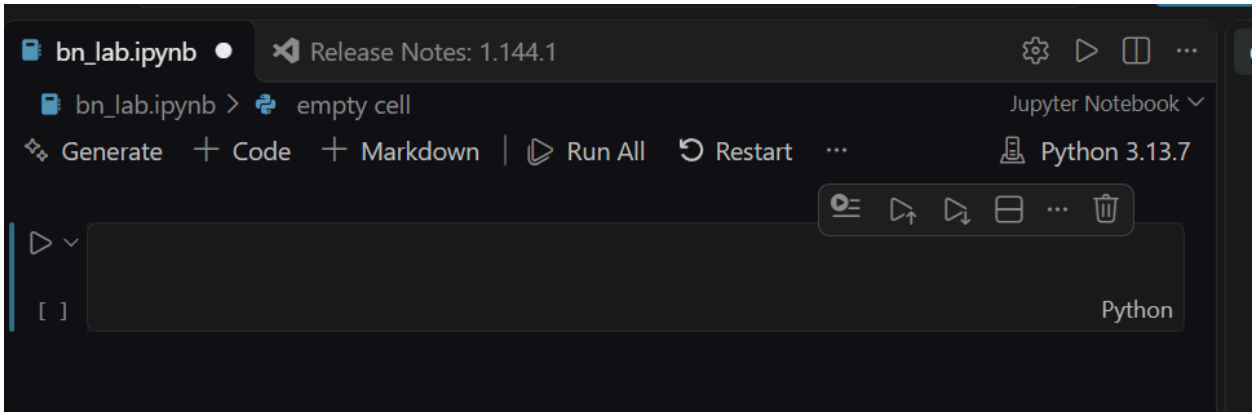


Step 2

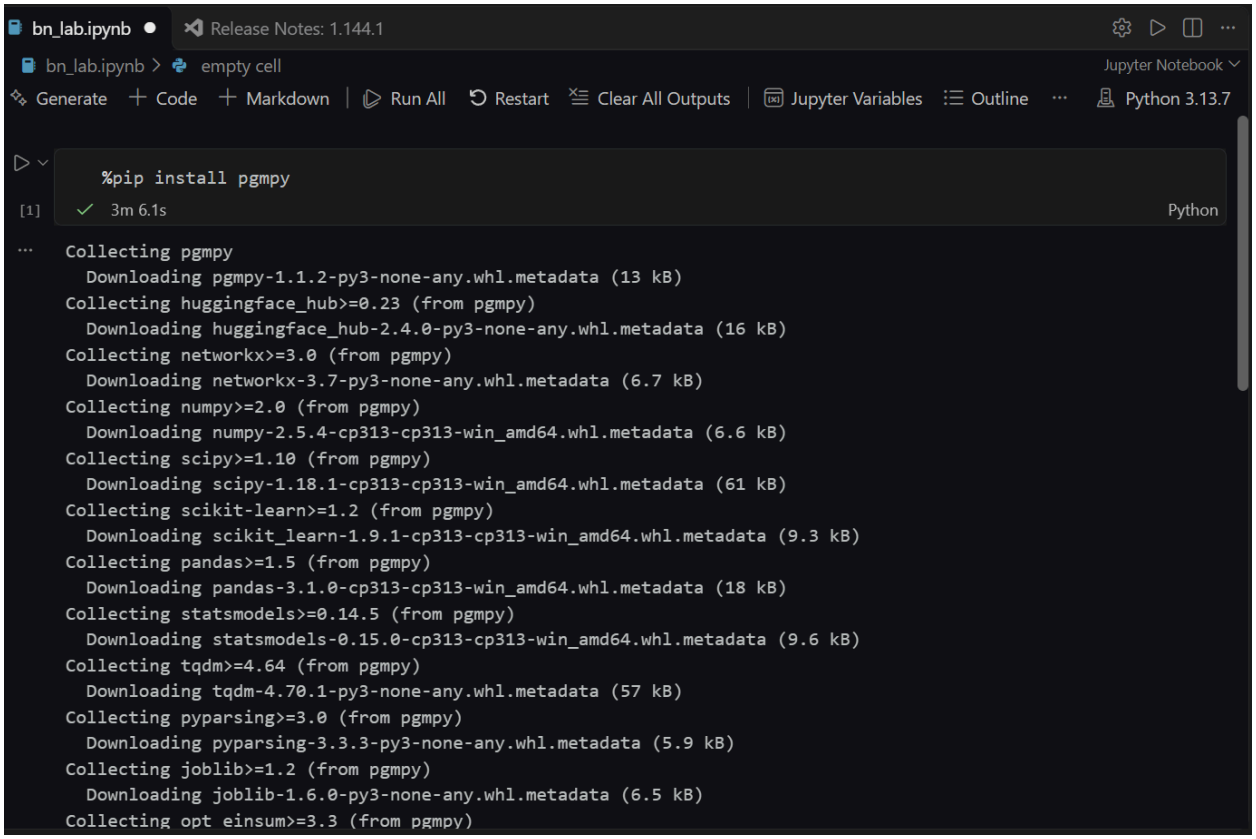




Step 3


bn_lab.ipynb
Release Notes: 1.144.1
bn_lab.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Python 3.13.7
[ ]
Python

Step 4


bn_lab.ipynb
Release Notes: 1.144.1
bn_lab.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
Python 3.13.7
%pip install pgmpy
[1]
3m 6.1s
Python
Collecting pgmpy
Downloading pgmpy-1.1.2-py3-none-any.whl.metadata (13 kB)
Collecting huggingface_hub>=0.23 (from pgmpy)
Downloading huggingface_hub-2.4.0-py3-none-any.whl.metadata (16 kB)
Collecting networkx>=3.0 (from pgmpy)
Downloading networkx-3.7-py3-none-any.whl.metadata (6.7 kB)
Collecting numpy>=2.0 (from pgmpy)
Downloading numpy-2.5.4-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Collecting scipy>=1.10 (from pgmpy)
Downloading scipy-1.18.1-cp313-cp313-win_amd64.whl.metadata (61 kB)
Collecting scikit-learn>=1.2 (from pgmpy)
Downloading scikit_learn-1.9.1-cp313-cp313-win_amd64.whl.metadata (9.3 kB)
Collecting pandas>=1.5 (from pgmpy)
Downloading pandas-3.1.0-cp313-cp313-win_amd64.whl.metadata (18 kB)
Collecting statsmodels>=0.14.5 (from pgmpy)
Downloading statsmodels-0.15.0-cp313-cp313-win_amd64.whl.metadata (9.6 kB)
Collecting tqdm>=4.64 (from pgmpy)
Downloading tqdm-4.70.1-py3-none-any.whl.metadata (57 kB)
Collecting pyparsing>=3.0 (from pgmpy)
Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Collecting joblib>=1.2 (from pgmpy)
Downloading joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting opt_einsum>=3.3 (from pgmpy)

```
  Downloading opt_einsum-3.4.0-py3-none-any.whl.metadata (6.3 kB)
Collecting scikit-base>=0.12.4 (from pgmpy)
...
   ---------------------------------------- 31/31 [pgmpy]

Successfully installed anyio-4.16.0 click-8.5.0 cloudpickle-3.1.2 filelock-4.2.7 formulaic-1.2.2 fsspec-2026.9.
Note: you may need to restart the kernel to use updated packages.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

[notice] A new release of pip is available: 25.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Python

Step 2

```
# 6) Put the tables into the graph and check them
model.add_cpds(rain, maintenance, train, appointment)
print("Model is correct:", model.check_model())
```

[1] ✓ 31.6s

```
Model is correct: True
```

Step 3

```
# P(r, m, t, a) = P(r) x P(m | r) x P(t | r, m) x P(a | t)
p = (rain.get_value(rain="none")
     * maintenance.get_value(maintenance="no", rain="none")
     * train.get_value(train="on time", rain="none", maintenance="no")
     * appointment.get_value(appointment="attend", train="on time"))
print(round(p, 4))
```

✓ 0.0s

```
0.3402
```

Step 4

```
p = (rain.get_value(rain="heavy")
     * maintenance.get_value(maintenance="yes", rain="heavy")
     * train.get_value(train="delayed", rain="heavy", maintenance="yes")
     * appointment.get_value(appointment="miss", train="delayed"))
print(round(p, 4))
```

✓ 0.0s

```
0.0024
```

Step 5

```
    show({"train": "delayed"})
4]  ✓ 4.2s

rain
    none: 0.4601
    light: 0.3005
    heavy: 0.2394
maintenance
    yes: 0.3662
    no: 0.6338
train: delayed   (observed)
appointment
    attend: 0.6000
    miss: 0.4000
```

Step 8

```
show({"appointment": "miss"})
```

[5] ✓ 0.0s

```
rain
    none: 0.6065
    light: 0.2392
    heavy: 0.1544
maintenance
    yes: 0.3441
    no: 0.6559
train
    on time: 0.4802
    delayed: 0.5198
appointment: miss   (observed)
```

Step 9

[6] ✓ 0.0s

```
rain: heavy   (observed)
maintenance
    yes: 0.1176
    no: 0.8824
train: delayed   (observed)
appointment
    attend: 0.6000
    miss: 0.4000
```

CS3081 Bayesian Network Lab: Likelihood and Inference

